Every well-ordering is isomorphic to a unique ordinal

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the minimal-counterexample shrink, the Replacement definition of f, and all three verification claims (range, ordering, ordinal domain) are present verbatim with their named internal citations.

Editorial note. The label thmOrdinalRepresentation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Uniqueness of ordinal representations. Same unit, \ollabel{ordisoidentity} (source line 599). Gives at-most-oneness at the head of the proof and makes the Replacement-built assignment f single-valued.
- Order comparison of ordinal-segment isomorphisms. Same unit, \ollabel{lemordsegments} (source line 820). At line 1045: two initial segments isomorphic to ordinals beta and gamma satisfy gamma in beta iff the induced map preserves ordering, so f is order-preserving.
- Transitive sets of ordinals are ordinals. Same unit, \ollabel{corordtransitiveord} (source line 534). At line 1047: reduces showing dom(f) is an ordinal to transitivity.
- Ordinals of isomorphic initial segments. Same unit, \ollabel{wellordinitialsegment} (source line 799). At line 1051: gamma in beta and beta in dom(f) force gamma in dom(f), closing transitivity of dom(f).
- ZF set theory with Replacement. The unit's standing background: the section header line 1018 records the author working in ZF minus Powerset, armed with Replacement. Replacement collects the per-segment isomorphism ordinals into the single set f at lines 1038-1042.

Source reference. Original corollary corordtransitiveord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 534–534. Prerequisite pointer invoked at proof line(s) 1047; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordisoidentity: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 599–599. Prerequisite pointer invoked at proof line(s) 1026; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lemordsegments: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 820–820. Prerequisite pointer invoked at proof line(s) 1045; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference wellordinitialsegment: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 799–799. Prerequisite pointer invoked at proof line(s) 1051; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Every well-ordering is isomorphic to a unique ordinal.*

Proof. Let $\langle A, <\rangle$ be a well-order. By ordisoidentity, it is isomorphic to at most one ordinal. So, for reductio, suppose $\langle A, <\rangle$ is not isomorphic to *any* ordinal. We will first "make $\langle A, <\rangle$ as small as possible". In detail: if some proper initial segment $\langle A_a, <_a\rangle$ is not isomorphic to any ordinal, there is a least $a \in A$ with that property; then let $B = A_a$ and $\lessdot = <_a$. Otherwise, let $B = A$ and $\lessdot = <$.

By definition, every proper initial segment of B is isomorphic to some ordinal, which is unique as above. So by Replacement, the following set exists, and is a function:

$$f = \{\, \langle \beta, b\rangle \mid b \in B \text{ and } \beta = \langle B_b, \lessdot_b\rangle \,\}$$

To complete the reductio, we'll show that f is an isomorphism $\alpha \to B$, for some ordinal α.

It is obvious that $\operatorname{ran} f = B$. And by lemordsegments, f preserves ordering, i.e., $\gamma \in \beta$ iff $f(\gamma) \lessdot f(\beta)$. To show that $\operatorname{dom} f$ is an ordinal, by corordtransitiveord it suffices to show that $\operatorname{dom} f$ is transitive. So fix $\beta \in \operatorname{dom} f$, i.e., $\beta = \langle B_b, \lessdot_b\rangle$ for some b. If $\gamma \in \beta$, then $\gamma \in \operatorname{dom} f$ by wellordinitialsegment; generalising, $\beta \subseteq \operatorname{dom} f$. □